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ORIGINAL RESEARCH

Enhanced lightweight infrared object detection algorithm for assistive navigation in visually impaired individuals

Zhimin Bai[1] | Yang Yang[2] | Jian Wang[1] | Zhengyang Li[1] | Jiajun Wang[1] | Chunxin Liu[1]

[1]School of Computer Science, North China Institute of Aerospace Engineering, Langfang, China

[2]Research Laboratory of Systems Software Engineering, Beijing Aerospace Automatic Control Institute, Beijing, China

Correspondence
Jian Wang, School of Computer Science, North China Institute of Aerospace Engineering, Langfang 065099, China.
Email: wjluky@126.com

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Abstract

This study introduces an advanced infrared scene detection algorithm, enhancing the YOLOv8 model for aiding visually impaired individuals in navigation. The focus is on the neck network, integrating attention scale sequences to boost multi-level perception, particularly for small object detection. This is achieved by adding upsampling and downsampling in the P2 module. Additionally, the CIoU loss function is refined with Inner-SIoU, elevating bounding box detection precision. A distinctive feature of the approach is its monocular distance and velocity measurement integration, which operates independently of external devices, providing direct navigation support for visually impaired people. Further, the enhanced YOLOv8 is adapted for mobile use, employing pruning and lightweight methods, which substantially enhance its practicality. The experimental results on the FLIR and WOTR datasets demonstrate that, compared to the original YOLOv8n, the improved algorithm has achieved a 2.1% and 3.2% increase in $mAP_{0.5}$, respectively. Furthermore, the $mAP_{0.5–0.95}$ has seen a 2.2% and 3.8% improvement. Concurrently, the model size has been reduced by 55% and 60%, and the number of parameters has decreased by 60% and 67%. Compared to other assistive travel methods for visually impaired individuals, our work demonstrates superior practicality.

1 | INTRODUCTION

With advancements in science and technology, sophisticated methodologies have been developed for object detection and positioning. These include the integration of ultrasonic sensors [1], light detection and ranging (LiDAR) [2], and infrared thermal imaging technologies [3]. Each of these systems offer unique advantages in accurately detecting and locating objects in various environments. In the realm of computer vision, the development of technologies to aid visually impaired individuals in navigation has emerged as a prominent research area within object detection. Leveraging deep learning, this technology facilitates real-time identification and alerts regarding pedestrians, non-motorized, and motorized vehicles in infrared environments. This enhancement not only expands the mobility range for the visually impaired but also significantly improves their daily life convenience. More importantly, it contributes to the safety of visually impaired travellers, mitigating the risk of accidents and ensuing injuries.

Infrared imaging [3] is a process that utilizes the emission, reflection, or conduction of infrared radiation by objects. This technology primarily exploits the radiative characteristics of the infrared spectrum, capturing this radiation with detectors and converting it into visible images. It has found wide applications in fields such as military, healthcare, security, environmental monitoring, and driver assistance. This method can provide valuable information in low-light or no-light conditions, thus offering particular advantages in specialized applications. In the context of assisting visually impaired individuals in navigation, infrared imaging technology presents a unique challenge. It demands rapid and accurate detection and identification of obstacles and objects in the environment, crucial for ensuring safe and effective navigation in an infrared scene.

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Recent academic research has significantly propelled the development of infrared scene object detection in the field. Inspired by the contrast mechanism of the human visual system, Chen et al. [4] introduced a compact object detection algorithm using a derivative kernel model. This approach stands out for its simplicity and effectiveness in detection accuracy. Liu et al. [5] developed an enhanced template matching technique for infrared object detection, demonstrating good performance in point object recognition. Li et al. [6] proposed a region-free infrared image detector, YOLO-FIR, based on the YOLOv5 architecture. Their method innovatively extended the cross-stage partial connection (CSP) module in the shallow cross-stage, thereby enhancing the utilization of shallow features and achieving impressive infrared object detection results. Furthermore, Fan et al. [7] presented an infrared vehicle detection algorithm utilizing an improved YOLOv5 framework. The algorithm employed dense connectivity blocks for superior feature extraction and integrated channel attention mechanism along with modified loss functions, resulting in improved overall network detection accuracy. Zhao et al. [8] significantly improved the detection accuracy of small infrared objects in resource-limited environments by combining residual structures, fusion-based feature reuse, and introducing DCA attention mechanism and MSIoU loss function. Despite achieving progress in speed and accuracy, many of these models still face challenges in deploying on small devices and often lack the required accuracy for detecting small objects.

In the field of assistive travel for visually impaired individuals, various studies have proposed targeted solutions to address specific challenges. Shen et al. [9] developed a wearable device for blind pedestrians that uses real-time object detection and shape-memory alloy (SMA) actuators to provide tactile feedback on obstacle positions. By reducing neural network redundancy by 90%, the device is suitable for low-memory mobile devices. However, it requires expensive auxiliary equipment. See et al. [10] created a smartphone-based mobility assistant using a depth camera for obstacle avoidance and object recognition, achieving over 80% accuracy with audio and vibration feedback. Despite not needing additional hardware, it generates constant alerts for up to 90 object types, lacking targeted category reduction. Islam et al. [11] introduced a low-cost assistive system using deep learning for obstacle detection and environment description. Employing TensorFlow, SSDLite, and Raspberry Pi, the system provides real-time audio feedback but uses outdated algorithms and does not handle sudden dangerous situations well. Kumar et al. [12] developed a navigational system using a YOLO-based algorithm for visually impaired individuals. The model, relying on the outdated YOLOv3 and only 300 training images, presents limitations in performance.

While deep learning-based techniques have significantly advanced object detection, surpassing human visual accuracy [13] in some cases, they are not without limitations. In complex real-world environments, these methods still face challenges such as missed detections and false alarms. Traditional detection methods often struggle with varying lighting conditions and handling smaller objects. This limitation becomes particularly evident in infrared imaging, where detecting small objects such as bicycles and pedestrians remains challenging. The inherent constraints faced by traditional approaches in such scenarios highlight the need for continuous progress in this field.

Assisting visually impaired individuals in object detection within infrared scenes presents notable differences from traditional object detection, primarily due to the following significant characteristics: (1) Objects are often densely distributed, leading to more frequent occlusions and overlaps. (2) Algorithms must adapt to various urban road environments, including complex background settings. (3) Considering the constraints of device deployment, computational capabilities, storage capacity, energy consumption, compactness, and portability must be taken into account. To address these challenges, this paper introduces a novel blind navigation detection algorithm that leverages tailored enhancements within the YOLOv8 model for the infrared scene. The main contributions and innovations of this work are outlined as follows:

1. Enhanced neck network: The attention scale sequence fusion [14] was applied to enhance the neck network, significantly improving the multi-level perceptual capability within the field of view.
2. Improvements in object detection: The P2 module in the neck has been enhanced with both upsampling and downsampling operations, specifically aimed at enhancing the detection accuracy of smaller objects.
3. Optimization of the CIoU [15] loss function: The original CIoU loss function has been optimized by incorporating Inner-SIoU [16], resulting in improved precision in bounding box detection.
4. Monocular distance and speed measurement: We have introduced a novel feature for real-time distance and speed measurement, eliminating the need for external devices and providing prompt warnings for visually impaired users.
5. Algorithm optimization for mobile deployment: The enhanced YOLOv8 algorithm has undergone pruning and lightweight processing, significantly reducing its size and parameter count. This renders it highly suitable for deployment on mobile platforms.

2 | MATERIALS AND METHODS

2.1 | YOLOv8 network architecture

YOLOv8 [17] represents a significant advancement in the field of object detection models, building upon its predecessor's high accuracy and rapid recognition capabilities. This model has widespread applications, supporting a range of tasks including object detection, image classification, and instance segmentation. In this study, we utilized YOLOv8 for object detection. YOLOv8 offers multiple configurations, namely YOLOv8n, YOLOv8s, YOLOv8m, YOLOv8l, and YOLOv8x, each with varying network depths and widths. Considering the requirements of our study and the need to strike a balance between

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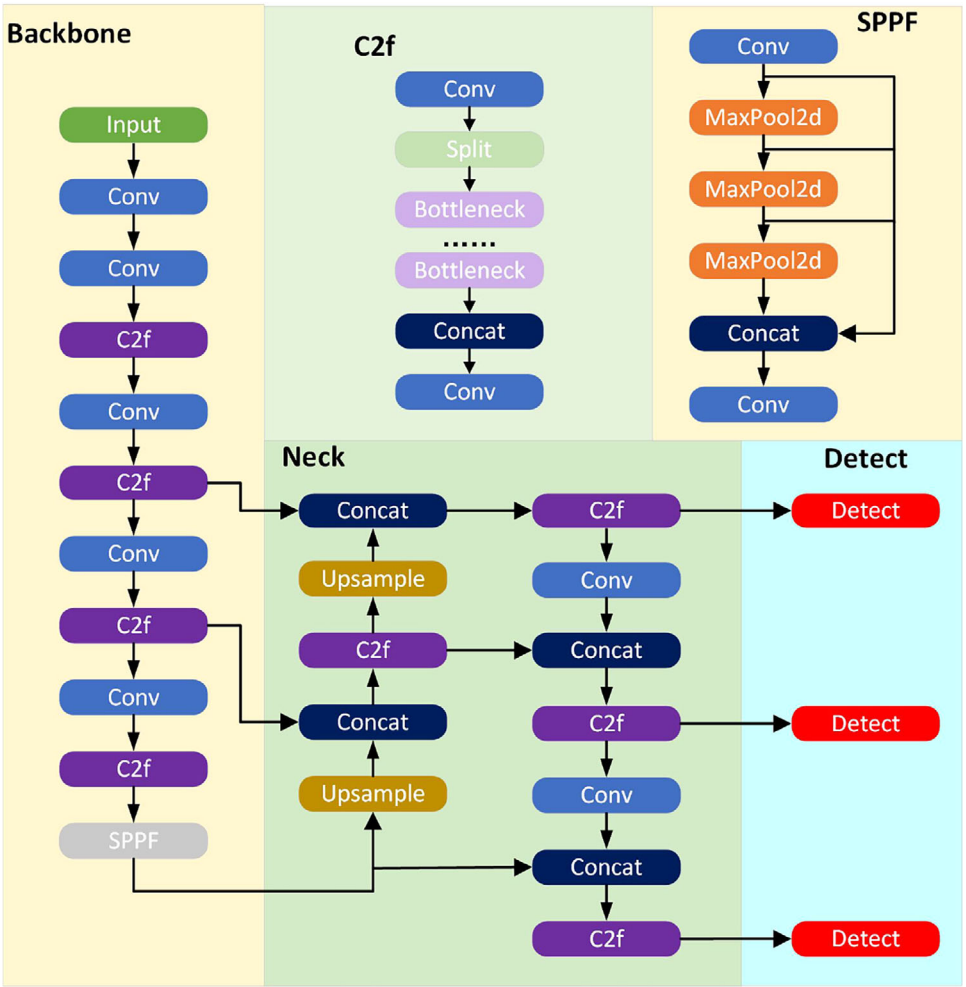


FIGURE 1 Network structure of YOLOv8.

model size and accuracy, we opted for the YOLOv8n variant. The YOLOv8n model is structured into four main components: input, backbone, neck, and head, as illustrated in Figure 1. This configuration has been optimized to effectively and efficiently detect objects in practical applications.

2.1.1 | Input

In the input stage, YOLOv8 employs mosaic data augmentation technique [18], similar to its predecessor YOLOv5. However, during the final 10 epochs, this augmentation technique is disabled to stabilize the training. It is noteworthy that YOLOv8 introduces a model-size-specific data augmentation intensity approach. It adjusts certain hyperparameters based on the model variant, recognizing that larger models may benefit from more dense augmentation strategies. For example, for larger YOLOv8 models, advanced techniques such as MixUp [19] and CopyPaste [20] are enabled. This customized data augmentation approach reflects a nuanced understanding of how different model architectures respond to different augmentation techniques.

2.1.2 | Backbone

The backbone component of YOLOv8 is composed of modules such as Conv, C2f, and SPPF, which play a crucial role in feature extraction. The Conv module primarily conducts convolutional operations, batch normalization (BN), and SiLU activation function processing on the input images. A significant improvement in YOLOv8 is the replacement of the C3 structure (seen in YOLOv5) with the C2f structure. This modification is of paramount importance as the C2f structure is optimized for richer gradient flow, making it the core module for learning residual features. This structural choice enables YOLOv8 to maintain a lightweight design while ensuring a rich gradient information flow. Additionally, the SPPF module [21] (i.e. spatial pyramid pooling) plays an indispensable role in transforming feature maps of different sizes into fixed-size

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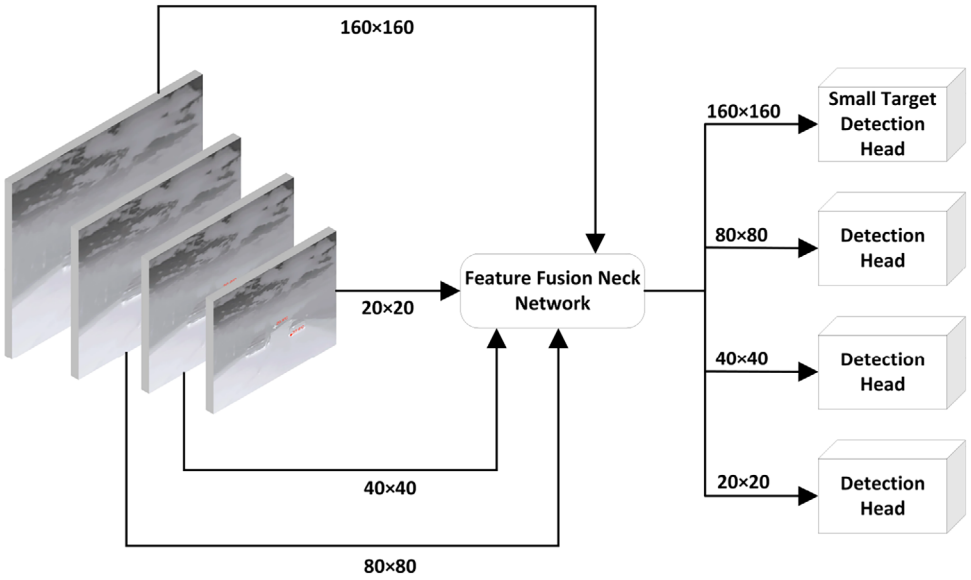


FIGURE 2 Principle of YOLOv8-P2.

feature vectors, further enhancing the model's adaptability and efficiency in handling inputs of varying dimensions.

2.1.3 | Neck

The neck component in the YOLOv8 architecture leverages the PANet [22] structure to generate a feature pyramid by integrating multi-scale features. This structure comprises two key elements: the feature pyramid network (FPN) [23] and the path aggregation network (PAN) [22]. FPN initially constructs the pyramid by extracting feature maps from the convolutional neural network. It then adopts a bottom-up approach for feature fusion, combining features from different levels with coarser feature maps. This process effectively reduces spatial resolutions while increasing feature channels. In contrast, PAN, designed to complement FPN, employs a top-down approach. It fuses feature maps from different levels through a convolutional layer, thus more accurately preserving spatial information. The synergistic interaction of FPN and PAN in the neck structure enhances the feature extraction and fusion capabilities of YOLOv8, enabling it to tackle complex object detection tasks effectively.

2.1.4 | Head

In comparison to YOLOv5, YOLOv8's head component has two significant improvements. Firstly, it separates the object classification and bounding box regression tasks, allowing for independent predictions. This separation helps accelerate model convergence speed and improve detection accuracy. In terms of classification, YOLOv8 introduces varifocal loss (VFL) [24], characterized by asymmetric weighting of positive and negative samples, emphasizing the former. This method effectively addresses the imbalance between positive and negative samples. For regression, the model adopts the distribution focal loss (DFL) module [25]. DFL enables the network to focus more quickly on the distribution near the actual object position.

The second major improvement is the adoption of Anchor-Free Detection (AFD) [26] instead of traditional anchor-based detection (ABD) [27]. AFD can directly predict the center point and aspect ratio of the objects without predicting the position and size of anchor frames. This transition reduces the number of anchor frames, thereby improving the model's detection speed and accuracy.

The detailed framework of the YOLOv8 is in Figure 1.

2.2 | Small object detection head

In the standard YOLOv8 model, the output layer consists of P3, P4, and P5. To enhance the detection of smaller objects, additional upsampling and downsampling operations are integrated into the neck component. This includes the inclusion of the P2 feature map, which increases the resolution by a factor of four, specifically for detecting very small objects. The P2 layer generates a larger feature map with a size of 160×160, specifically for detecting objects larger than 4×4. This improvement greatly enhances the model's ability to identify small-sized objects.

While the output from the backbone remains unchanged, modifications are made to the structure of the neck and head components to accommodate the additional feature extraction capability for smaller objects. Figure 2 provides a detailed description of the modified model structure after the inclusion of the P2 layer for enhanced detection of small objects.

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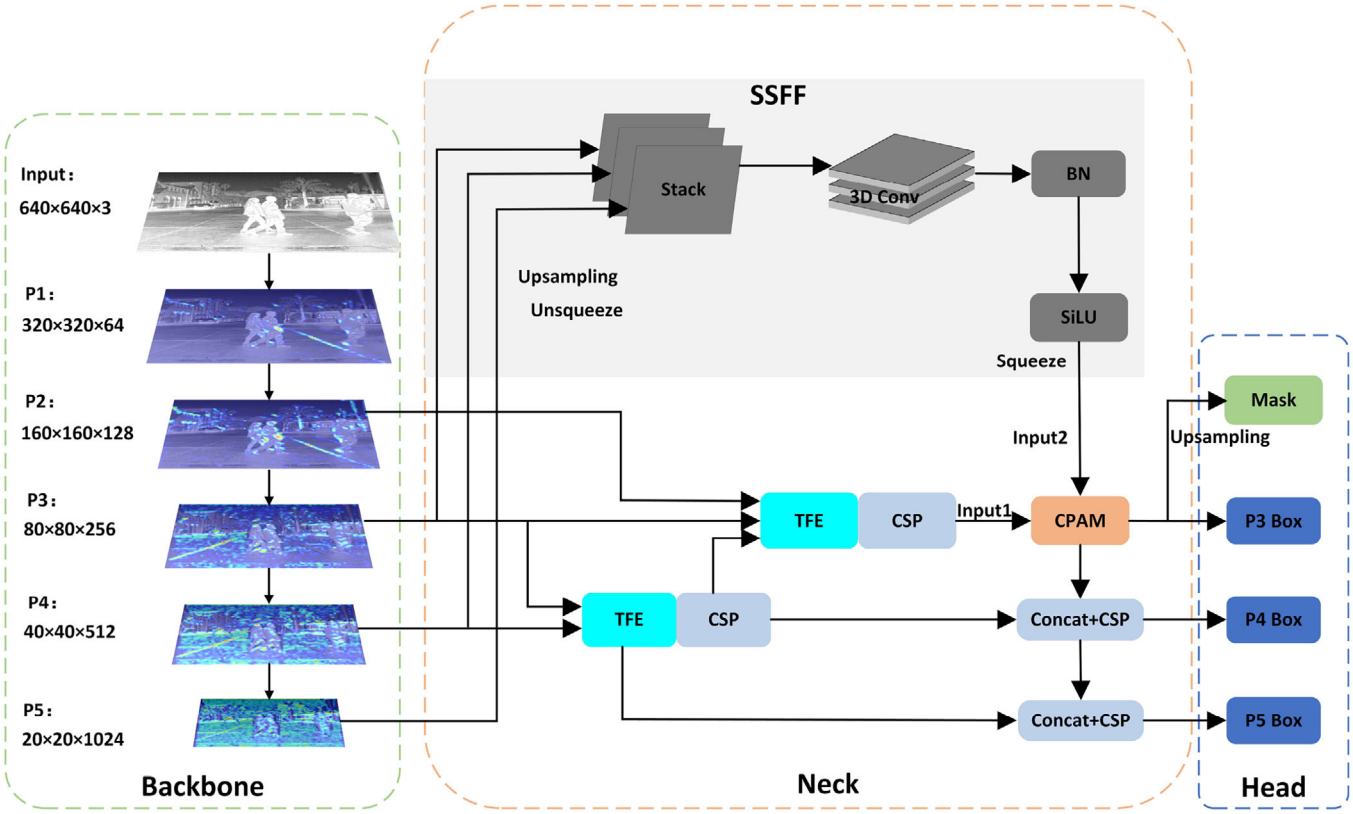


FIGURE 3 Network structure of YOLOv8-ASF.

2.3 | Attentional scale sequence fusion

In the neck network of YOLOv8, the initial fusion step involves combining the output features of P3 and P4 from the backbone network. Simultaneously, these features undergo spatial scale feature fusion (SSFF) [14] with the output features of P5. This process begins by expanding the dimensions of these three feature sets to three dimensions. They are then normalized to a uniform size and stacked, effectively fusing the features captured by P3, P4, and P5 with different spatial scales, sizes, and shapes. Subsequently, these fused features undergo a series of three-dimensional operations: convolution, batch normalization, activation, and max pooling. These operations effectively capture information at different spatial scales, ultimately restoring the output to two-dimensional information.

Additionally, the network implements the object feature enhancement (TFE) [14] fusion operation. TFE receives three different feature maps of different scales, the original P2, P3, and the fused features of P3 and P4. It then integrates these different features into a comprehensive intermediate-scale feature map, enhancing the overall feature representation of the model. The network structure of ASF-YOLOv8 is illustrated in Figure 3.

The TFE operation in the YOLOv8 neck network involves different processing steps for feature maps of different scales. For large feature maps, TFE first applies convolution, batch normalization, and an activation function, doubling the number of channels. The resulting feature map is then downsampled and subjected to max and average pooling. Upon merging the pooling outputs, the size of the feature map is halved. Subsequently, another round of convolution, batch normalization, and activation is carried out to maintain the channel number and size.

Conversely, small feature maps processed with TFE increase the number of channels, followed by upsampling using the nearest neighbour method, effectively doubling its scale. It then undergoes convolution, batch normalization, and an activation function, without changing the channel number or size.

Medium-sized feature maps only require one TFE operation and retain their channel number and size. This is achieved through convolution, batch normalization, and an activation function.

The final result of these processes is three feature maps with uniform channel number and size. These feature maps are then stacked to merge multi-scale features. Finally, the TFE-fused features are integrated with the output of spatial scale feature fusion (SSFF), fully leveraging the integrated advantages of advanced multi-scale and detailed features. The network structure of TFE is shown in Figure 4.

2.4 | YOLO-ASFP

In this study, we adopted YOLOv8 as the base model and introduced specific improvements to enhance its performance. The key enhancement lies in the neck component, which

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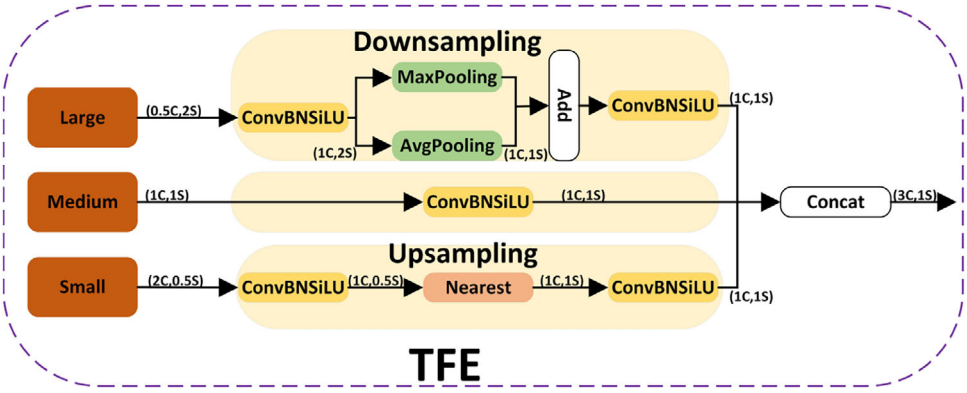


FIGURE 4 Principle of the TFE module.

integrates the output features of the backbone network P2, P3, P4, and P5 during the upsampling stage. This fusion effectively consolidates prominent features extracted from the backbone network, enriching the feature representation. Additionally, we incorporated upsampling and feature fusion operations during the downsampling stage of the neck. This modification enables the network to capture a more comprehensive set of features in the final output through a single detection head.

The head component of the network is configured to output four detection heads, with each detection head specifically designed to identify objects of different sizes. The algorithm integrates attention scale sequence fusion and small object detection techniques. This dual approach enables the model to capture more crucial features, thereby improving overall detection accuracy. Importantly, this design also enhances the model's ability to detect small objects, addressing the common issue of missing small objects in detection models. The network structure of YOLOv8-ASFP is shown in Figure 5.

2.5 | Inner-SIoU loss

In order to ensure the accurate localization of bounding boxes when conducting object detection using YOLOv8, especially for subsequent calculations involving the size of bounding boxes, refining the loss function is crucial. To minimize the difference between the predicted values and the ground truth, the refinement of the loss function is necessary. The original YOLOv8 utilizes the complete intersection over union (CIoU) algorithm to calculate the regression loss between the predicted box and the ground truth box. The CIoU loss function is defined as follows:

$$L_{\mathrm{CIoU}} = 1 - \mathrm{IoU} + \frac{\rho^2(b, b^{\mathrm{gt}})}{C^2} + \alpha v \tag{1}$$

$$\alpha = \frac{v}{(1 - \mathrm{IoU}) + v} \tag{2}$$

$$v = \frac{4}{\pi^2}\left(\arctan\frac{w^{\mathrm{gt}}}{h^{\mathrm{gt}}} - \arctan\frac{w}{h}\right)^2 \tag{3}$$

Where $\rho^2(b, b^{\mathrm{gt}})$ represents the Euclidean distance between the centers of the predicted bounding box and the ground truth bounding box. C represents the diagonal distance of the smallest closed region that can simultaneously contain both the predicted bounding box and the ground truth bounding box. w, h, w^{gt}, h^{gt} denote the widths and heights of the predicted bounding box and the ground truth bounding box, respectively.

Although CIoU effectively represents the overlapping area between predicted and ground truth bounding boxes, it does not consider the directional consistency between the predicted and ground truth bounding boxes, which can lead to the drift effect in the predicted frames. This often results in suboptimal performance of the model. Additionally, the standard IoU [28] metric is challenging to accurately detect small objects, leading to limitations in CIoUs precise bounding box regression for these objects.

To address the issues with CIoU, this paper introduces the SCYLLA-IoU (SIoU) [29] loss function. SIoU consists of four components: angle cost, distance cost, shape cost, and IoU cost, the SIoU loss function is defined as follows:

$$L_{\mathrm{box}} = 1 - \mathrm{IoU} + \frac{\Delta + \Omega}{2} \tag{4}$$

$$\mathrm{IoU} = \frac{|B \cap B^{\mathrm{GT}}|}{|B \cup B^{\mathrm{GT}}|} \tag{5}$$

$$\Omega = \sum_{t=w,h}(1 - e^{-\omega_t})^{\theta} \tag{6}$$

$$\omega_w = \frac{|w - w^{\mathrm{gt}}|}{\max(w, w^{\mathrm{gt}})}, \omega_h = \frac{|h - h^{\mathrm{gt}}|}{\max(h, h^{\mathrm{gt}})} \tag{7}$$

$$\Delta = \sum_{t=x,y}(1 - e^{-\gamma\rho_t}) \tag{8}$$

$$\rho_x = \left(\frac{b^{\mathrm{gt}}_{c_x} - b_{c_x}}{C_w}\right)^2, \rho_y = \left(\frac{b^{\mathrm{gt}}_{c_y} - b_{c_y}}{C_h}\right)^2, \gamma = 2 - \Lambda \tag{9}$$

$$\Lambda = 1 - 2 * \sin^2\left(\arcsin(x) - \frac{\pi}{4}\right) \tag{10}$$

$$x = \frac{c_h}{\sigma} = \sin(\alpha) \tag{11}$$

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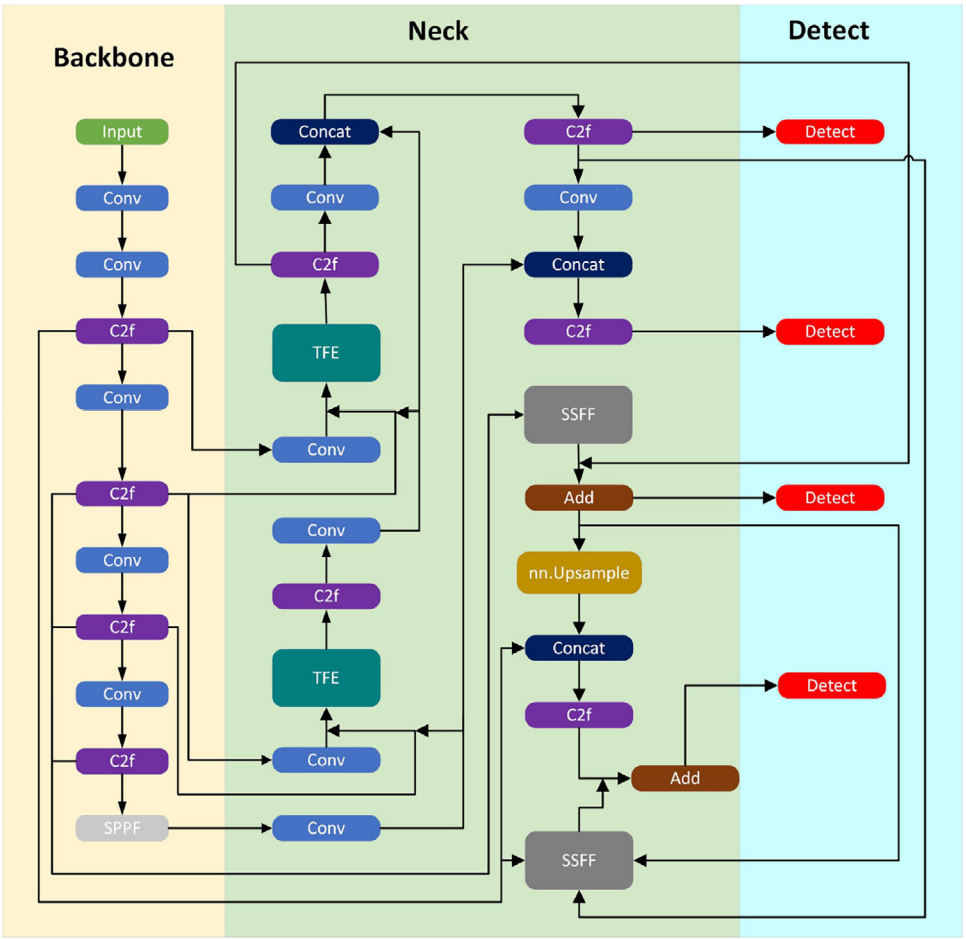


FIGURE 5 Network architecture of YOLOv8-ASFP.

$$\sigma = \sqrt{\left(b_{c_x}^{gt} - b_{c_x}\right)^2 + \left(b_{c_y}^{gt} - b_{c_y}\right)^2} \tag{12}$$

$$c_h = \max\left(b_{c_y}^{gt}, b_{c_y}\right) - \min\left(b_{c_y}^{gt}, b_{c_y}\right) \tag{13}$$

Where w^{gt}, h^{gt}, w, and h represent the width and height of the ground truth and predicted bounding boxes; b_{cx}^{gt}, b_{cy}^{gt}, b_{cx}, and b_{cy} denote the coordinates of the centers of the ground truth bounding box and the predicted bounding box, respectively; C_w and C_h represent the width and height of the minimum bounding rectangle of the ground truth bounding box and the predicted bounding box; c_h denotes the height difference between the centers of the ground truth bounding box and the predicted bounding box. σ is the distance between the center point of the real frame and the predicted frame.

SIoU introduces a more sophisticated bounding box regression method, which includes perspective factors and size sensitivity. This method effectively overcomes the limitations of the CIoU loss function, can accelerate convergence speed, and improve the model's accuracy. However, SIoU does not completely address the inherent challenges of the IoU loss itself, which may affect detection quality and model generalization.

To further improve the model's accuracy and generalization ability, this paper combines inner-IoU with SIoU and ultimately uses the inner-SIoU loss function to predict bounding box loss. The inner-IoU loss function is defined as follows:

$$b_l^{gt} = x_c^{gt} - \frac{w^{gt} * \text{ratio}}{2}, b_r^{gt} = x_c^{gt} + \frac{w^{gt} * \text{ratio}}{2} \tag{14}$$

$$b_t^{gt} = y_c^{gt} - \frac{h^{gt} * \text{ratio}}{2}, b_b^{gt} = y_c^{gt} + \frac{h^{gt} * \text{ratio}}{2} \tag{15}$$

$$b_l = x_c - \frac{w * \text{ratio}}{2}, b_r = x_c + \frac{w * \text{ratio}}{2} \tag{16}$$

$$b_t = y_c - \frac{h * \text{ratio}}{2}, b_b = y_c + \frac{h * \text{ratio}}{2} \tag{17}$$

$$\text{inter} = \left(\min\left(b_r^{gt}, b_r\right) - \max\left(b_l^{gt}, b_l\right)\right) * \left(\min\left(b_b^{gt}, b_b\right) - \max\left(b_t^{gt}, b_t\right)\right) \tag{18}$$

$$\text{union} = \left(w^{gt} * h^{gt}\right) * (\text{ratio})^2 + (w * h) * (\text{ratio})^2 - \text{inter} \tag{19}$$

$$\text{IoU}^{\text{inner}} = \frac{\text{inter}}{\text{union}} \tag{20}$$

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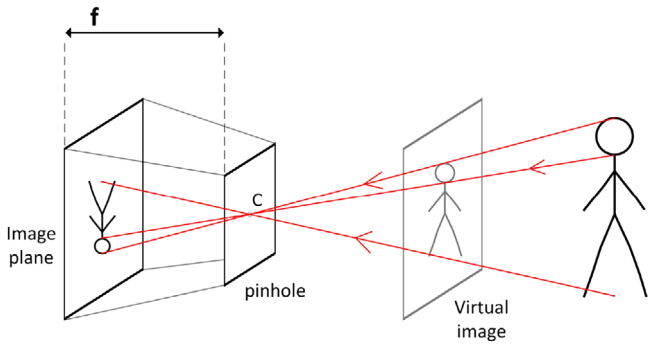


FIGURE 6 Principle of single camera rangefinding.



FIGURE 7 Demonstration of speed and distance measurement video images captured using infrared vehicle recorder.

Where B_{gt} and B respectively represent the ground truth (actual) bounding box and the anchor (predicted) bounding box. The centres of the ground truth bounding box and the inner truth box are denoted as (x_c^{gt}, y_c^{gt}), while (x_c, y_c) represents the center of the anchor box and its corresponding inner box. The dimensions of the ground truth bounding box are represented by width and height, denoted as w^{gt} and h^{gt}, while the dimensions of the anchor bounding box are denoted as w and h. The ratio is a scale factor, typically ranging from 0.5 to 1.5. When the ratio is less than 1, the auxiliary box size is smaller than the actual box, and its regression has a smaller effective range than the IoU loss. However, the absolute value of its gradient is larger than the gradient obtained from the IoU loss, which can accelerate the convergence of high IoU samples. Conversely, when the ratio is larger than 1, the larger scales of the auxiliary box expand the effective range of regression, providing gains for regression with low IoU. The inner-SIoU loss function is defined as follows:

$$L_{\text{Inner-SIoU}} = L_{\text{SIoU}} + \text{IoU} - \text{IoU}^{\text{inner}} \tag{21}$$

Inner-SIoU is an innovative loss function that combines the advantages of inner-IoU and SIoU. This combination takes into account the angular difference between the actual bounding box and the anchoring box, as well as the intrinsic properties of ionic power unit losses, resulting in a comprehensive loss assessment.

2.6 | Distance measurement

The principle of single-camera ranging is illustrated in Figure 6, where f represents the focal length of the camera, and c is the optical center of the lens. The light from the object travels through the camera's optical center and is captured on the image sensor, or, in other words, is formed on the image plane. With this principle we can get the distance calculation formula:

$$D = \frac{F * W}{P} \tag{22}$$

In Equation (22), D represents the distance from the object to the camera (i.e. the distance of pedestrians, non-motorized vehicles, and motorized vehicles), F is the focal length of the camera, W is the width or height of the object (i.e. the true height of pedestrians, non-motorized vehicles, and motorized vehicles), and P is the width of the object in the x-direction pixels or the height of the object in the y-direction pixels (which can be obtained from the object detection results of YOLOv8).

To perform distance measurement using a monocular camera, the first step is to set the lens focal length, which can be obtained from the specifications of the recording device. Next, the actual height of pedestrians, non-motorized vehicles, and motorized vehicles must be determined in inches. These parameters are then used in the distance calculation formula mentioned above to easily compute the distance to these entities. This process essentially maps real-world dimensions to pixel dimensions on the image sensor, enabling accurate distance estimation.

When measuring vehicle speed, the first step is to gather data from the current frame and the previous frame. This data includes the centroid coordinates (x and y) of each detected vehicle, the object identification number, the vehicle category, and the pixel width. The next step is to calculate the pixel displacement between the centroid points of vehicles in these two frames. Then, the pixel displacement is converted to actual distance using a linear mapping equation. To determine the distance travelled by the vehicles in real space, this distance is divided by the width of the vehicles. Finally, the movement distance is divided by the time interval between frames to calculate the velocity.

However, it is crucial to address a common issue where the measured velocity underestimates the actual velocity, especially when the camera's field of view is not perpendicular to the vehicle's trajectory. To mitigate this problem, an adjustment factor (typically around 2) needs to be applied to make the measured velocity closer to the actual velocity. Figure 7 demonstrates a video image example of speed and distance measurement using an infrared driving recorder.

2.7 | Prune

To accelerate the inference process while maintaining model effectiveness, we have adopted modern strategies such as

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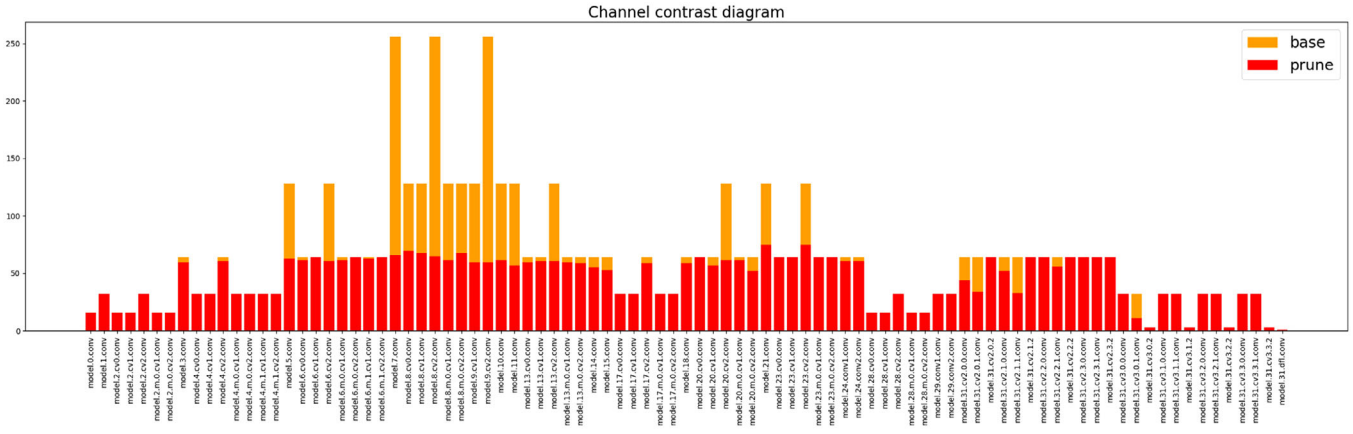


FIGURE 8 Comparison of changes in the number of channels before and after model pruning.

pruning, quantization, and knowledge distillation. Although quantization is effective, its deployment often requires specialized hardware support, which may not be feasible in embedded systems. On the other hand, knowledge distillation relies on the ability of the student model to absorb and retain the knowledge from the teacher model, a process that typically demands significant computational resources and time.

In contrast, pruning techniques offer a more accessible solution as they do not require complex computations or specific hardware. This technique selectively removes redundant weights from neural networks, as these weights are deemed unimportant for model accuracy. Pruning can be performed in a structured or unstructured manner, each objecting different aspects of network architecture. By enhancing the sparsity of object detection models through pruning, we can effectively balance model complexity and performance, making it a viable choice for various applications, including those with hardware constraints.

Layer-wise adaptive magnitude pruning (LAMP) [30] is a global pruning method for neural networks that autonomously determines the sparsity level of each layer. This approach discards traditional heuristic methods or complex hyperparameter adjustments. LAMP uses a unique metric, termed the LAMP score, to measure the importance of weights within the network. This score represents a recalibrated version of weight magnitudes designed to reflect the impact of pruning on the overall model performance. The calculation formula for the LAMP score is:

$$\text{score}(u; W) = \frac{(W[u])^2}{\sum_{v \geq u}(W[v])^2} \tag{23}$$

In Equation (23) $W[u]$ represents the weight value at the index u in the sorted weight list for a given layer. This formula essentially quantifies the importance of each connection in the network, aiming to preserve the connections that minimize overall distortion in the pruned model's output.

This computation aims to quantify the significance of each neural connection, with the goal of preserving the connections that are crucial for minimizing distortion in the pruned model's output. By prioritizing the retention of these important connections, LAMP ensures that the loss of model accuracy is minimized while improving efficiency.

TABLE 1 Comparison of pruning rates on FLIR datasets.

Pruning rate/%	$mAP_{0.5}$/%	$mAP_{0.5–0.95}$/%	Mode size/MB
40	**86.9**	**51.5**	3.8
50	86.6	51.4	2.7
60	84.2	50.2	2.2
70	80.5	48.7	**1.8**

The LAMP score is crucial for quantifying the importance of each neural connection in the quantized network. It is calculated by evaluating the magnitude of weights, comparing each weight with others within their respective layers. A higher LAMP score indicates a greater impact on the model's output. By prioritizing the retention of weights with higher LAMP scores during the pruning process, this method effectively minimizes the adverse effects on model performance.

One of the main advantages of the LAMP method is its ability to provide an assessment of weight importance for specific layers, thus improving the precision and efficiency of the pruning process. This feature makes LAMP particularly advantageous in scenarios where reducing model size while maintaining performance is desirable. Figure 8 illustrates the comparison of channel numbers between the original model and the corresponding pruned model, demonstrating the effectiveness of the LAMP method in optimizing model architecture.

The pruning process of the enhanced model includes strategically eliminating redundant channels, effectively reducing the total number of parameters and streamlining the model. As a result, the model becomes more lightweight without compromising its core functionality, aligning with the objectives of efficient computation and resource utilization reduction.

To optimize the pruning process and determine the most effective pruning ratio that does not excessively degrade the model's capabilities, we tested a series of pruning ratios, including 40%, 50%, 60%, and 70%. The corresponding results are detailed in Tables 1 and 2. Observations reveal that increasing

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TABLE 2 Comparison of pruning rates on WOTR datasets.

Pruning rate/%	$mAP_{0.5}$/%	$mAP_{0.5–0.95}$/%	Mode size/MB
40	**75.1**	**48.9**	2.5
50	74.6	48.5	2.2
60	71.3	44.0	1.7
70	68.5	40.7	**1.4**

TABLE 3 Comparison of results of different pruning methods on FLIR datasets.

Pruning methods	$mAP_{0.5}$/%	$mAP_{0.5–0.95}$/%	Mode size/MB
LAMP	**86.6**	**51.4**	**2.7**
slim	85.3	49.7	3.3
group_slim	85.5	49.8	3.1
group_norm	86.0	50.2	2.8
group_sl	84.3	48.6	2.9
group_taylor	83.5	47.6	3.0

TABLE 4 Comparison of results of different pruning methods on WOTR datasets.

Pruning methods	$mAP_{0.5}$/%	$mAP_{0.5–0.95}$/%	Mode size/MB
LAMP	**74.6**	**48.5**	**2.2**
slim	71.2	43.8	2.3
group_slim	72.8	43.7	3.3
group_norm	73.4	43.1	2.7
group_sl	71.4	45.2	3.2
group_taylor	70.3	39.8	2.9

the pruning ratio leads to varying degrees of decline in detection accuracy. At a pruning ratio of 40%, the model achieves the highest detection accuracy, but also has the largest size. Conversely, at higher pruning ratios (60% and 70%), although the model size significantly decreases, the detection accuracy also noticeably declines. A pruning ratio of 50% is a balanced choice as it effectively maintains a relatively high detection accuracy while reducing the model's size.

To validate the effectiveness of the proposed LAMP pruning model, we compared it with five different pruning frameworks: slim [31], group_slim [32], group_norm [33], group_sl [33], and group_taylor [34]. The comparison results are shown in Tables 3 and 4. The enhanced LAMP pruning model has achieved an average precision (mAP) of 86.6% and 74.6% on two separate datasets, respectively. This performance indicates its remarkable ability to detect small objects. Moreover, the model's size effectively reduces to 2.7 and 2.2 MB, meeting the requirement for lightweight deployment. Compared to other pruning methods, the LAMP model achieves an excellent balance between detection efficiency and model size.

TABLE 5 Software and hardware platform configuration.

Environment	Configure
GPU	NVIDIA Geforce RTX4090
CPU	AMD EPRC 7T83@2.5 GHz
Memory	90GB
Memory (GPU)	24GB
Operating system	Ubuntu20.04
Frame work	Python3.8 PyTorch1.13.1
CUDA	11.8

TABLE 6 Hyper-parameter settings.

Super parameter	Values
Epochs	200
Patience	50
Batch	32
Imgsize	640
Optimizer	SGD
Momentum	0.937
Initial learning rate	0.01
Final learning rate	0.0001

3 | EXPERIMENT AND ANALYSIS

3.1 | Experimental environment

The hardware and software platform configuration parameters for this experiment are shown in Table 5. The parameters are set as shown in Table 6.

3.2 | Dataset

This study utilized both the FLIR [35] and Walk On The Road (WOTR) [36] datasets. The FLIR dataset, publicly accessible and released by FLIR in 2018, comprises 10,288 infrared images. These images have been annotated to identify three common mobility obstacles for visually impaired individuals: pedestrians, bicycles, and vehicles. The WOTR dataset contains approximately 190,000 objects, with an average of 13.6 objects per image. It includes 15 categories of common obstacles and 5 categories for road assessment, covering various complex road scenarios. This dataset is used to validate the effectiveness of our improved model on a multi-object category dataset. We have provided annotated data on the https://github.com/Jackbai360/labels website to support further academic research in this field. Both datasets are split into training, testing, and validation sets in a ratio of 7:2:1.

In addition to annotated samples, the dataset also includes "negative" sample images that do not contain any specified categories. These images are accompanied by blank TXT label files

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TABLE 7 Sample classification criteria.

Classification	Instruction
TP	Number of correctly categorized positive samples
FP	Number of positive samples misclassified
TN	Number of negative samples correctly categorized
FN	Number of negative samples misclassified

and are included in the training process. The inclusion of these unannotated samples is a deliberate measure aimed at enhancing the robustness of the dataset and ensuring its conformity to real-world everyday scenarios. This approach helps reduce potential misclassifications in the model, thus improving the overall accuracy and reliability of the model. This methodology serves to minimize potential misclassifications within the model, thereby enhancing the overall accuracy and reliability of the model.

3.3 | Model evaluation metrics

This experiment evaluates the model using precision, recall, average precision AP and average precision mAP. The sample classification criteria are shown in the Table 7:

(1) Precision

Object detection accuracy is defined as the likelihood that a sample recognized as positive by the model is indeed positive.

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{24}$$

It is calculated as the ratio of the sum of the true positive (TP) predictions for samples correctly identified as positive to the false positive (FP) predictions for samples incorrectly identified as positive.

(2) Recall

Recall measures the ability of the model to correctly predict positive samples from all actual positive instances.

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{25}$$

This is calculated by dividing TP by the sum of TP and false negative (FN) instances, where the model incorrectly predicts positive samples as negative.

(3) AP (average precision)

Average precision (AP) is the average precision value at different levels of recall and essentially represents the average classification performance of a single category. It is quantified by the area under the precision–recall (P–R) curve in the first quadrant.

$$\mathrm{AP} = \int_0^1 P(R)\mathrm{d}|R \tag{26}$$

Higher AP values indicate fewer detection errors.

(4) mAP (mean average precision)

Mean accuracy (mAP) is the set of AP values in different categories, reflecting the precision of the model within a certain range of recall values.

$$\mathrm{mAP} = \frac{1}{n}\sum_{i=1}^{n}(\mathrm{AP})_i \tag{27}$$

The higher the mAP value, the better the model's ability to achieve both high precision and high recall.

3.4 | Visualization results

To demonstrate the enhanced capability of the improved YOLOv8n algorithm in detecting small objects, visual comparison results between the original YOLOv8n algorithm and the enhanced version are presented in Figures 9 and 10. These images emphasize the detection of small objects such as cars, bicycles, and pedestrians. Clear distinctions in the performance of these two models are evident, with the original YOLOv8n algorithm exhibiting a relatively high rate of missed detections for these small objects. In contrast, the improved YOLOv8n algorithm shows a significant reduction in missed detections and a noticeable improvement in detection accuracy. This comparison distinctly illustrates the advanced capability of the improved algorithm in identifying small objects, which is a crucial aspect of the performance of object detection models.

Figures 9a and 9b present the comparison of detection results on the test set before and after applying improvements to the YOLOv8n model. The improved results demonstrate a significant enhancement in the model's ability to capture global information and recognize a broader range of categories. There is also a noticeable increase in the confidence associated with the identified object categories. Particularly noteworthy is the model's improved capability in detecting smaller objects such as pedestrians and bicycles. This improvement can be attributed to the integration of ASF (attention scale sequence fusion) and the inclusion of the P2 module, which is specifically designed for small object detection. In contrast, the original YOLOv8n model exhibits a higher frequency of missed detections for these smaller objects.

Heatmaps are crucial visual tools for object detection, used to describe the intensity distribution of detected objects within input images. These heatmaps aid in precisely locating object positions and displaying their confidence, with brighter areas indicating higher model detection confidence. Figures 10c and 10d intuitively illustrate the comparison between the base model and the enhanced YOLOv8 model using heatmaps. These images clearly demonstrate that the improved YOLOv8 model not only has a more precise recognition range, but also exhibits higher detection confidence. This comparison emphasizes the enhanced model's outstanding capability in accurately identifying and confidently determining object detection results.

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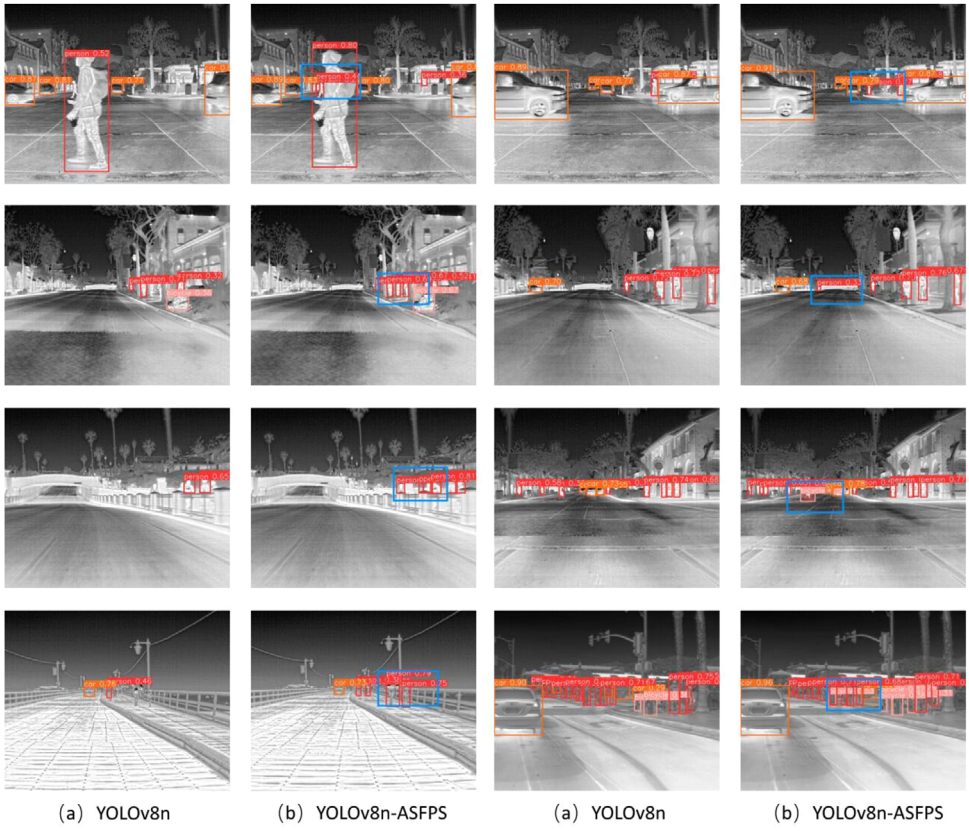


FIGURE 9 Comparison of YOLOv8n and YOLOv8n-ASFPS detection results on FLIR datasets.

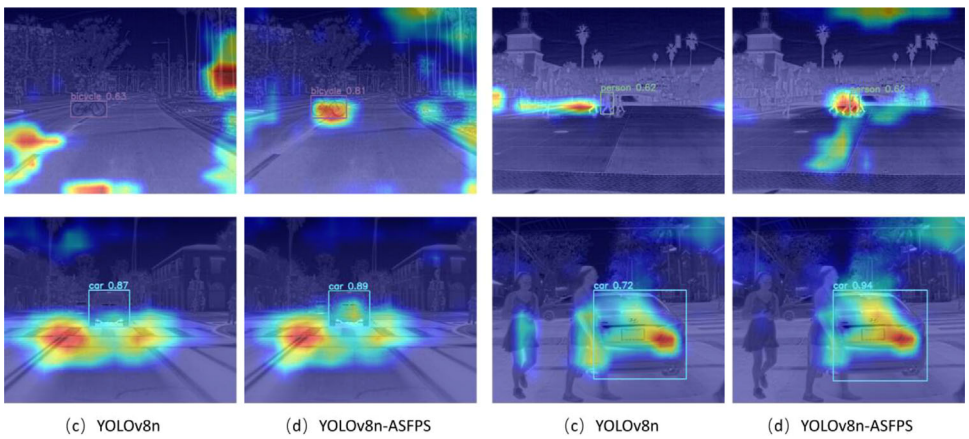


FIGURE 10 Comparison between heatmaps of YOLOv8n and YOLOv8n-ASFPS.

3.5 | Ablation experiments

To validate the effectiveness of the proposed model improvements, we conducted ablation studies. These experiments aimed to maintain consistency in model parameters across different runs to ensure the reliability of the results. Tables 8 and 9 comprehensively present the findings of these studies. This approach allows for a comprehensive evaluation of the specific enhancements to the model and their impact on the overall performance in the field of object detection.

Figure 11 presents a comparative analysis of the model metrics after integrating various modules, with detailed results presented in Table 8. The results of the ablation studies indicate a significant improvement in model accuracy after enhancing the neck network of YOLOv8 by incorporating scale sequence fusion, enabling the model to effectively assimilate multi-level

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TABLE 8 Ablation experiments on FLIR datasets.

+ASF	+P2	+Inner-SIoU	+prune	$mAP_{0.5}$/%	$mAP_{0.5–0.95}$/%	Params/M	Flops/G	Size/MB
				84.6	49.2	3.0	8.1	6.0
✓				85.9	51.0	3.04	8.5	6.0
	✓			86.7	51.6	2.92	12.2	6.0
		✓		85.1	49.8	3.0	8.1	6.0
✓	✓			86.3	50.8	2.48	12.0	5.1
✓	✓	✓		**87.2**	**51.5**	2.48	12.0	5.1
✓	✓	✓	✓	86.6	51.4	**1.20**	**7.9**	**2.7**

TABLE 9 Ablation experiments on WOTR datasets.

+ASF	+P2	+Inner-SIoU	+prune	$mAP_{0.5}$/%	$mAP_{0.5–0.95}$/%	Params/M	Flops/G	Size/MB
				71.4	44.7	3.0	8.1	6.0
✓				72.6	45.2	3.04	8.5	6.0
	✓			73.7	46.6	2.92	12.2	6.0
		✓		72.2	45.4	3.0	8.1	6.0
✓	✓			74.4	47.0	2.48	12.0	5.1
✓	✓	✓		**75.9**	**49.5**	2.48	12.0	5.1
✓	✓	✓	✓	74.6	48.5	**0.98**	**7.9**	**2.2**

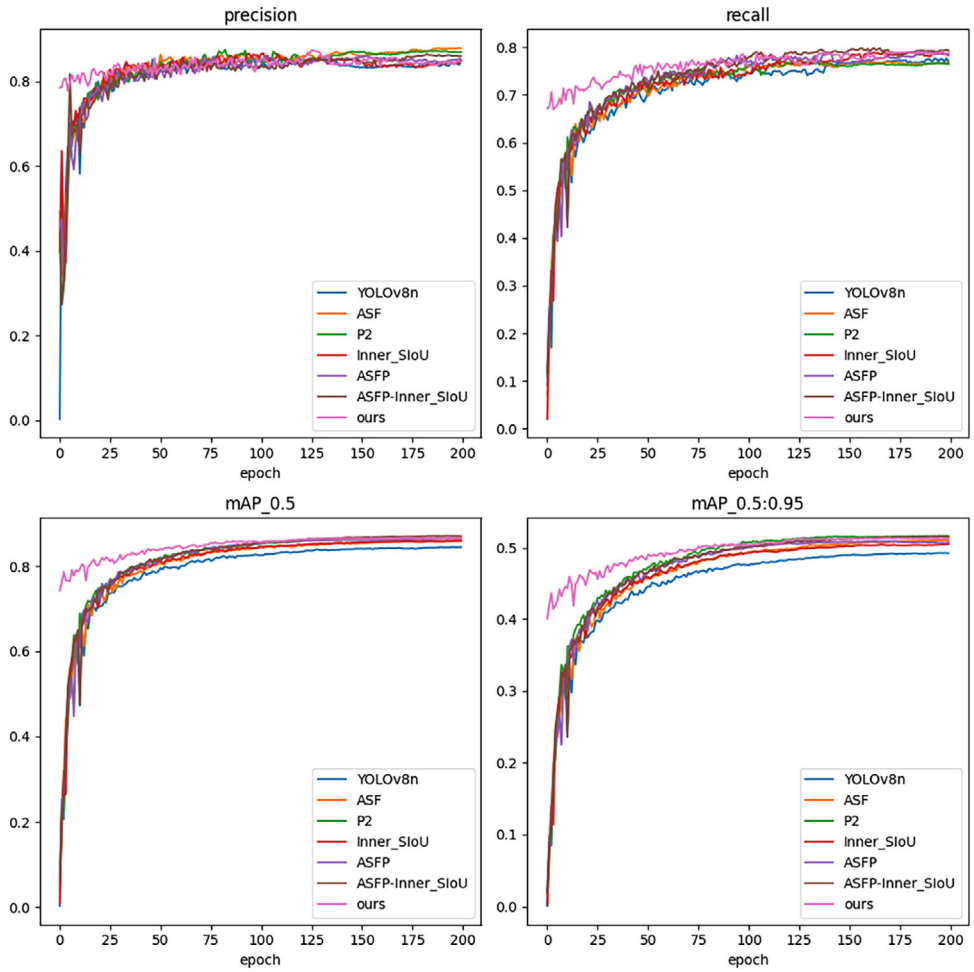


FIGURE 11 Comparison of results of various ablation experiments.

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TABLE 10 Comparison of mainstream object detection algorithms on FLIR datasets.

Model	$mAP_{0.5}$/%	$mAP_{0.5–0.95}$/%	Params/M	Flops/G	Size/MB
Faster R-CNN	84.2	47	41.13	206.67	315
SSDlite	39.3	26	3.06	7.03	23.7
YOLOv3	67	36	61.53	193.89	469
YOLOX-nano	69.3	37.6	**0.91**	**3.19**	11.1
YOLOv5n	84.5	46.1	1.76	4.1	3.7
YOLOv7-tiny	84.7	45.9	6.0	13.2	12.3
YOLOv8n	84.6	49.2	3.0	8.1	6.0
YOLO-ASFPS	**86.6**	**51.4**	1.20	7.9	**2.7**

TABLE 11 Comparison of mainstream object detection algorithms on WOTR datasets.

Model	$mAP_{0.5}$/%	$mAP_{0.5–0.95}$/%	Params/M	Flops/G	Size/MB
Faster R-CNN	73.6	45.2	41.13	206.67	315
SSDlite	37.9	26.7	3.06	7.03	23.7
YOLOv3	64.2	35.4	61.53	193.89	469
YOLOX-nano	66.9	38.2	**0.91**	**3.19**	11.1
YOLOv5n	73.7	44.4	1.76	4.1	3.7
YOLOv7-tiny	74.2	47.2	6.0	13.2	12.3
YOLOv8n	71.4	44.7	3.0	8.1	6.0
YOLO-ASFPS	**74.6**	**48.5**	0.98	7.9	**2.2**

information. Furthermore, the inclusion of the P2 network with four detection heads significantly improved the model's capability to detect small objects.

Compared to using the P2 module alone, the initial integration of ASF and the P2 module resulted in a slight performance decrease, which was effectively alleviated by the subsequent integration of inner-IoU. Overall, compared to the original model, the improved method achieved increases of 2.6% and 4.5% in $mAP_{0.5}$, and 2.3% and 4.8% in $mAP_{0.5–0.95}$, respectively. Additionally, this integration also contributed to reducing the model size and parameter count, highlighting the effectiveness of these comprehensive improvements in optimizing the task of object detection.

Following pruning, the model underwent channel reduction consistent with the convolutional layers of the backbone network to reduce computational demands. Specifically, at the 0.5 threshold, there was a 0.5% and 1.5% decrease. However, the significant improvement in model efficiency compensated for this reduction: the overall model size was reduced by 47% and 50%, and the parameter count decreased by 56% and 63%, respectively. Considering these modifications in conjunction with the aforementioned four improvements, it is evident that the enhanced model not only retained the accuracy of the original YOLOv8n model but also largely surpassed the original model. Furthermore, the improved model demonstrated a more compact and efficient nature in terms of parameter count and size, providing a more optimized solution for the task of object detection.

3.6 | Comparison of test results

To determine the comparative advantages of the proposed algorithm in this study, we conducted experimental analyses on several widely recognized detection models. The selected models for comparison include Faster R-CNN [37], SSDlite [38], YOLOv3 [39], YOLOX-nano [40], YOLOv5n [21], YOLOv7-tiny [41], and YOLOv8n [17]. These comparisons were performed under the same conditions, ensuring consistent configurations and using the same dataset to maintain experimental consistency. The evaluation metrics employed in this comparison include $mAP_{0.5}$, $mAP_{0.5–0.95}$, parameter count, FLOPS (floating-point operations per second), and model size. This set of comprehensive metrics provides an overall view of the performance of each model, including accuracy, computational efficiency, and resource utilization, enabling a comprehensive evaluation of the enhanced algorithm developed in this research.

Tables 10 and 11 provide a comparative analysis of various detection models, including the enhanced YOLOv8 model. YOLOX-nano exhibits the lowest parameter count and FLOPS, but its detection accuracy is relatively lower, surpassing only SSDlite and YOLOv3 in this aspect. On the other hand, the

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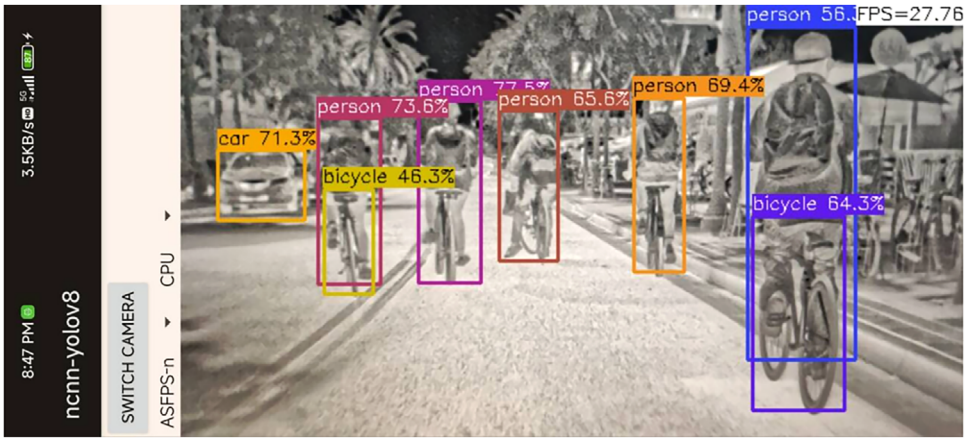


FIGURE 12 Detection effect of YOLOv8n-ASFPS on a smartphone.

improved YOLOv8 model demonstrates outstanding performance in several key metrics ($mAP_{0.5}$, $mAP_{0.5–0.95}$, and model size), surpassing all other comparison models. Additionally, its parameter count is second only to YOLOX-nano. These comparative experiments highlight the effectiveness of the enhanced YOLOv8 model in complex infrared scenes and varying object sizes, emphasizing its overall advantages in terms of accuracy, efficiency, and compactness.

3.7 | Validation of YOLO-ASFPS model for embedded deployment

We encountered computational limitations when deploying the latest improved YOLOv8 algorithm on edge computing devices such as Raspberry Pi and K210. On these platforms, the real-time frame rate achieved by the algorithm was only 5–10 FPS. Without leveraging devices like the intel neural compute stick, this level of performance is insufficient to achieve real-time object detection and danger warning, which are crucial functionalities for assisting visually impaired individuals in navigation. To meet the demand for a lightweight, portable, and cost-effective blind walking solution, we shifted our focus to smartphone platforms. By deploying the enhanced YOLOv8 model as a mobile application, we leverage the existing hardware capabilities of smartphones, thereby minimizing additional costs for users. Compared to traditional edge computing devices, modern smartphones have a significant advantage in terms of floating-point computational performance, thanks to advanced processing technologies and efficient power management. Therefore, we chose the Snapdragon 865 Mobile Platform as the deployment medium, striking a balance between performance requirements and practical constraints of visually impaired assistive technology.

NCNN [42] (abbreviation of Nocow network neural network) is a compact neural network computing framework optimized for mobile and embedded systems. Developed by Tencent YouTu Lab, NCNN is customized for high-performance neural network inference on different hardware platforms, including ARM processors and Mali GPUs. It facilitates the conversion of trained neural network models into proprietary binary formats and provides an inference library, enabling the execution of computationally intensive tasks such as real-time object detection on mobile devices.

NCNN allows for the deployment of lightweight deep learning models for real-time image processing and computational tasks, especially in resource-constrained environments like mobile and embedded systems. As shown in Figure 12, the final detection results prove the effectiveness of this approach. The deployed model achieved an average frame rate of 30 FPS, effectively meeting the real-time detection requirements in resource-limited environments.

4 | RESULTS AND DISCUSSIONS

4.1 | Software design and verification

The software for navigation assistance specifically designed for the visually impaired is intended to test the practical applicability of the algorithm. During the software startup and initialization phase, the system immediately performs a critical self-check process—network connection status detection. This step ensures that the software can intelligently allocate resources based on the current network environment. If the device is connected to the Internet, the software interfaces with the advanced speech synthesis and recognition API from iFlytek and the AutoNavi map API, leveraging the cloud's powerful data processing and real-time update capabilities to provide users with seamless, accurate voice interaction and navigation services. Conversely, if the device is offline, the software automatically loads pre-stored offline voice and map packages to ensure the continuity and availability of basic functions, demonstrating the software's flexibility and robustness. The design concept and implementation process of the software are illustrated in Figure 13.

After initialization and basic configuration, the software activates the camera, initiating a real-time visual recognition

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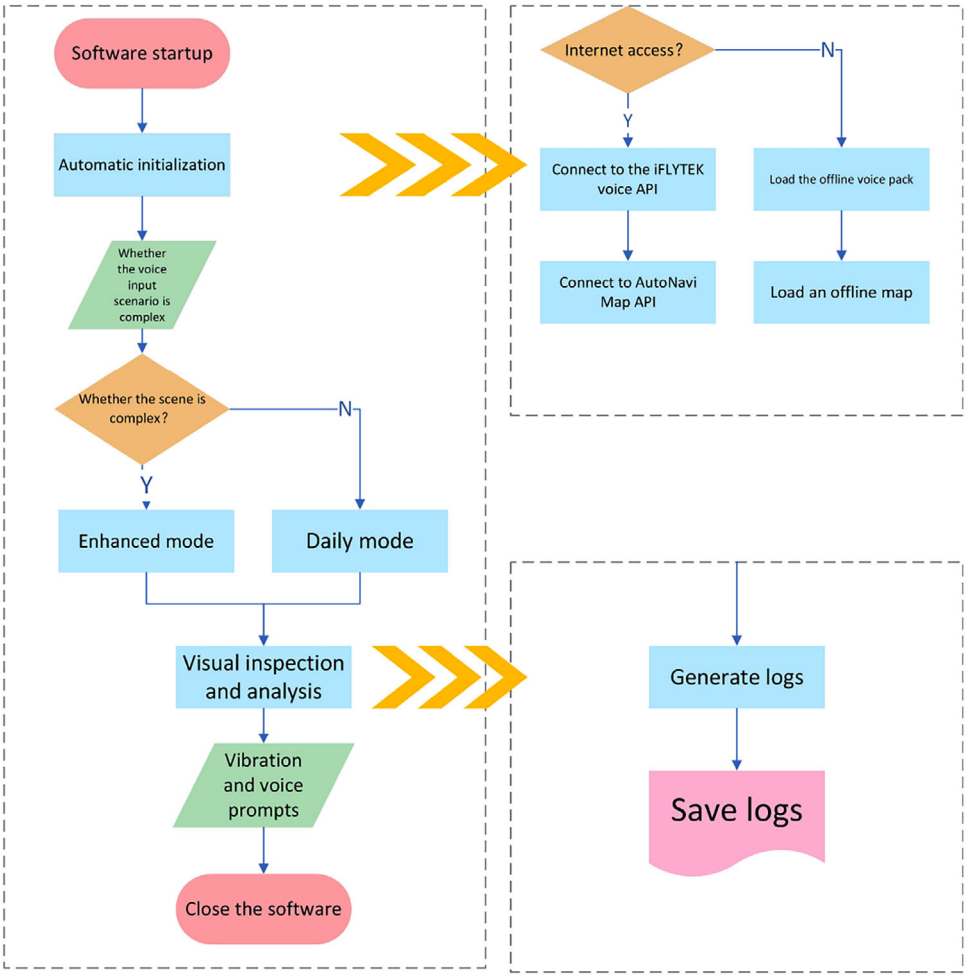


FIGURE 13 Software flow chart.

and analysis process. This process employs image processing technology to accurately assess the presence of obstacles in the user's surrounding environment. Upon detecting potential threats, such as approaching objects or unnoticed steps, the software immediately provides clear voice prompts and timely vibration feedback to ensure the safety of visually impaired users, effectively preventing accidents. The experimenter conducting a field simulation test on the street is shown in Figure 14.

If the system determines that the current scene is complex and dynamic, such as a crowded commercial area or a busy street, the software will automatically switch to enhanced visual assistance mode. This mode utilizes advanced computer vision algorithms to deeply analyse the images captured by the camera, identifying and marking potential obstacles, pedestrian flow directions, and other critical information to provide more comprehensive environmental awareness support. In contrast, for simpler daily travel scenarios, such as familiar neighbourhoods or parks, the software remains in daily mode, optimizing resource utilization to ensure ease of operation. Additionally, all recognition results and user interaction records are automatically logged, providing valuable data for subsequent feature optimization and personalized services.

4.2 | Comparison to other works

To validate the effectiveness of our research methodology, we compared it with methods proposed by other researchers, as shown in Table 12. In the domain of using deep learning to assist visually impaired individuals in detecting and recognizing objects, most related studies employ models such as YOLOv3 and SSD, trained on the COCO [48] dataset. These models are eventually deployed on wearable assistive devices to help visually impaired individuals detect and recognize objects. However, these models are relatively outdated and trained on conventional object detection datasets, which are not specifically tailored to scenarios faced by visually impaired individuals. Additionally, the reliance on assistive devices presents drawbacks, as these devices are costly and require users to undergo a learning process.

Moreover, most of the aforementioned studies lack the capability for speed measurement and distance estimation. While visually impaired individuals can more easily perceive static obstacles, they have difficulty detecting and reacting to suddenly approaching moving objects, such as vehicles, which poses a significant danger. Another critical issue is that overly complex datasets with a large variety of recognizable categories can

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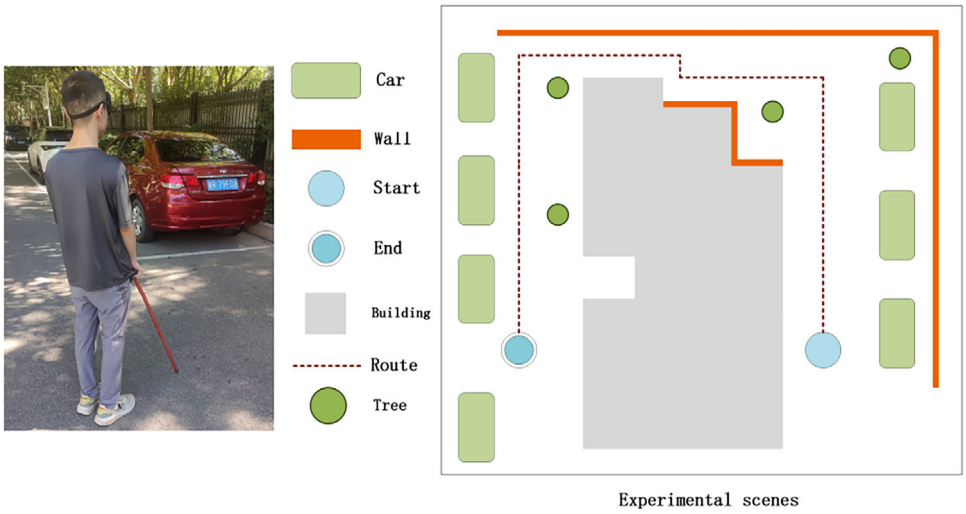


FIGURE 14 The experimenter conducted field simulations using a smartphone equipped with the software.

TABLE 12 Comparison to other works.

Reference	Primary features	Deployment platform	Applied model	Performance metrics	Limitations
[9]	Instantaneous recognition of objects with haptic feedback	Raspberry Pi 3B+	YOLOv3	Precision = 0.79 Recall = 0.72 Recognize obstacle accuracy = 96%	Requires additional hardware and is not portable
[10]	Smartphone equipped with depth camera for obstacle evasion	Smartphone	SSD MobileNet v2	N/A	Contains a significant amount of redundant and irrelevant object recognition categories
[11]	Barrier identification and environment description	Raspberry Pi 4B	SSDLite MobileNetV2	Precision = 0.89 Recall = 0.89 Detection accuracy = 88.89%	Model accuracy is constrained by the size of the weather dataset
[12]	Detection of items for enhanced navigation	Raspberry Pi 3B	YOLOv3	Detection accuracy = 81%	The algorithm exhibits low efficiency and lacks multimodal support
[43]	Identification and sorting of traffic signals	Raspberry Pi	OpenCV	Detection accuracy = 87.5%	The algorithm has low efficiency and a low accuracy rate
[44]	Pedestrian crossing detection for safe guidance	AI Edge MCI module	Inception SSD	Detection accuracy = 92.2%	Excessive external devices impede its portability
[45]	Barrier detection and traffic signal categorization	GD32F1 MCU	YOLOv3-tiny	mAP = 97%	Operates based on WiFi connectivity
[46]	Crosswalk identification for navigating safely	Raspberry Pi 4	OpenCV	Detection accuracy = 84.5%	The device is complex and not user-friendly
[47]	Animal recognition and categorization	Jetson Nano	SSD ResNet-50	mAP = 93.5%	Confined to the identification and detection of animal species
This work	Enhancement algorithm in infrared scene	Smartphone	Improved YOLOv8	mAP = 86%	Constrained by the battery life and camera capture limitations of mobile devices

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interfere with the mobility of visually impaired individuals, making it difficult for them to discern which alerts are of high priority, thus impeding their travel.

To address these issues, we have implemented targeted improvements. We utilized the state-of-the-art YOLOv8 model and enhanced its accuracy in detecting small objects. Additionally, we integrated monocular tracking and speed measurement functionalities into the model to provide real-time alerts for various types of objects. Our model has undergone pruning and lightweight processing, making it highly suitable for deployment on mobile platforms. We trained our model using the FLIR and WOTR datasets, which are specifically designed for visually impaired individuals, eliminating many irrelevant categories. Finally, we deployed our lightweight model on a smartphone, achieving an average frame rate of 30 FPS, enabling real-time detection. This deployment eliminates the need for additional assistive devices, greatly facilitating the mobility of visually impaired individuals.

During our evaluation, certain limitations of the current methodology have been observed. Specifically, the battery life constraints of smartphones preclude the possibility of continuous, round-the-clock support. Furthermore, the camera's field of view, when held in hand, is restricted to the frontal aspect, thereby limiting the capability to detect and provide alerts for the surrounding environment in a full 360-degree panorama. It is imperative to refine and enhance the algorithm to extend its applicability to a wider range of scenarios.

5 | CONCLUSIONS

This paper presents an improved algorithm based on YOLOv8 for assisted navigation detection in infrared scenes, combining higher accuracy with lightweight deployment suitable for real-time applications. The key advancement lies in integrating the attention scale sequence fusion into the neck network, promoting multi-scale feature fusion and enabling comprehensive learning across different informational levels. By adjusting the number of channels in the convolutional layers of the backbone network, the computational load of the model has been optimized. Furthermore, the utilization of the P2 module in the neck facilitates upsampling and downsampling operations, significantly enhancing the detection capability for smaller targets. Additionally, the adoption of the inner-SIoU loss function improves the accuracy of target category detection, enhancing the overall performance of the model. These modifications collectively ensure the model's suitability for assisting visually impaired navigation, meeting the dual requirements of real-time detection and practical deployment.

In the context of assisting blind individuals in navigation, this paper extends the functionality of the improved model to include monocular tracking and speed measurement, providing real-time alerts for various target categories. In our experiments utilizing the FLIR and WOTR datasets, the enhanced version of YOLOv8n exhibited a $mAP_{0.5}$ increase of 2.1% and 3.2%, and a $mAP_{0.5–0.95}$ improvement of 2.2% and 3.8%, respectively. Additionally, model efficiency was notably improved, with a reduction in model size by 55% and 60%, and a decrease in parameter count by 60% and 67%. Deployed on mobile devices through the NCNN framework, the model achieves a real-time frame rate of 30 frames per second, presenting a novel and effective solution for assisting visually impaired navigation. Looking ahead, we plan to further optimize this algorithm to adapt to more complex urban environments, expanding its applicability.

AUTHOR CONTRIBUTIONS

Zhimin Bai: Formal analysis; methodology; writing—ori-ginal draft. **Yang Yang**: Project administration; validation. **Jian Wang**: Supervision. **Zhengyang Li**: Data curation; visualization. **Jiajun Wang**: Visualization; writing—review & editing. **Chunxin Liu**: Conceptualization; validation.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available. FLIR dataset is available at [35] and WOTR dataset [36] is available at [49].

ORCID

Jian Wang https://orcid.org/0009-0006-1051-1359
Jiajun Wang https://orcid.org/0009-0006-1228-3361

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